

The Observer's Distortion Coefficient: Resolving the Hubble Tension and Galactic Rotation Curves via Golden Ratio Information Packing and Nested Relativistic Latency

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Date: July 2026

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Abstract

Modern cosmology and quantum mechanics are constrained by structural inconsistencies, requiring arbitrary physical constructs such as Dark Matter, Dark Energy, and wave-function collapse to reconcile continuous mathematics with empirical observations . This paper presents a unified digital physics framework, modeling physical reality as a discrete computational rendering execution loop running on a non-local grid geometry (the NavMesh). By enforcing an information-theoretic data-packing constraint derived from the Golden Ratio (ϕ) , we extract the universe's hardware base clock speed, the Server Tick ($\aleph_c \approx 8.72282 \times 10^{-19}$ seconds) .

We formulate a Unified Rendering Equation (R_{conso}) that integrates global expansion and local general relativity as nested computational latencies , deriving the terrestrial observer's processing rate ($R_{conso_Earth} \approx 8.71109 \times 10^{-19}$ seconds) . Based on this framework, we resolve the Hubble Tension as an algorithmic artifact of localized clock drift and redefine galactic rotation curves as a "Lazy Evaluation" optimization below the floating-point resolution floor (a_0).

Finally, we propose a definitive, falsifiable experiment: measuring the discrepancy of quantum "negative delay" time-stamps between Earth's surface and orbit (ISS), presenting a direct challenge to the global physics community to test this model's quantitative predictions.

1. Introduction: The Grid and Hardware Constraints

Discrete spacetime models often struggle with macroscopic Lorentz invariance violations. This framework bypasses continuous infinities by treating spacetime as an optimized Graphical User Interface (GUI), rendered by an underlying Base Operating System (Base OS). Particles are not independent physical objects but pointer variables to a non-local backend database (D-Dimension) , and mass is fundamentally a dense data payload requiring local CPU cycles to unpack on the discrete coordinate grid (NavMesh)) [1], [2].

2. Information Packing and the Base Server Tick ($\aleph_c$)

To prevent data collisions and spatial overlap during matrix translation, the Base OS utilizes a non-periodic, optimized coordinate distribution algorithm governed by the Golden Ratio ($\phi \approx 1.6180339887$) . We establish that the number of discrete processing frames consolidated into a single macro-spatial batch is governed by this mathematical limit. Utilizing the international baseline

for Planck time ($t_p \approx 5.391247 \times 10^{-44}$ seconds) , we define the unburdened hardware clock speed, or base Server Tick ($\aleph_c$):

$$\begin{aligned}\aleph_c &= t_p \times (\phi \times 10^{25}) \approx (5.391247 \times 10^{-44} \text{ s}) \times (1.6180339887 \times 10^{25}) \\ &\approx 8.72282 \times 10^{-19} \text{ s} \quad (1)\end{aligned}$$

Multiplying this hardware clock interval by the speed of light (c , representing the maximum horizontal data transmission bandwidth of the GUI) yields the spatial batch-processing resolution boundary (D):

$$\begin{aligned}D &= c \times \aleph_c \approx (299,792,458 \text{ m/s}) \times (8.72282 \times 10^{-19} \text{ s}) \\ &\approx 2.61503 \times 10^{-10} \text{ m} \quad (2)\end{aligned}$$

This spatial length ($2.615 \text{ \AA}$) corresponds precisely to the empirical characteristic atomic diameter of the most thermodynamically stable baryonic structures, such as Iron (^{56}Fe), representing the default, zero-compression data-packing block of the GUI.

3. The Unified Rendering Equation and Nested Latency (R_{conso})

Time is not a fundamental independent dimension, but an emergent local loading delay (Latency) arising from nested computational overhead. The Consolidated Processing Rate (R_{conso}) is modified concurrently by global cosmological expansion and local gravitational mass density:

$$R_{conso} = \aleph_c \times F_{global} \times F_{local} = \aleph_c \times \sqrt{1 - \left(\frac{\rho_b}{\rho_\Omega}\right)^2} \times \sqrt{1 - \frac{2GM}{rc^2}} \quad (3)$$

3.1 Global Cosmological Friction (F_{global})

The background global latency is determined by the ratio of baryonic payload ($\rho_b \approx 4.93\%$, representing high-overhead compressed data arrays) to the static, uncompressed vacuum density ($\rho_\Omega \approx 95.07\%$):

$$F_{global} = \sqrt{1 - \left(\frac{\rho_b}{\rho_\Omega}\right)^2} \approx 0.998654545 \quad (4)$$

3.2 Local Gravitational Latency and Kinematic Latency (F_{local})

Localized concentrations of mass-energy (data density) and high-velocity coordinate translations across the NavMesh (grid update rates) concurrently drain the server's processing budget. We refine the local latency factor (F_{local}) as a product of both gravitational and kinematic overhead:

$$F_{local} = F_{grav} \times F_{kin} = \sqrt{1 - \frac{2GM}{rc^2}} \times \sqrt{1 - \frac{v^2}{c^2}} \quad (5)$$

Where G is the gravitational constant, M is the localized mass, r is the radial coordinate distance, and v is the velocity of the observer relative to the unburdened background grid.

An observer on Earth is subject to both latencies, yielding the friction-adjusted macroscopic rendering clock speed (R_{conso_Earth}) [2]:

$$R_{conso_Earth} = (8.72282 \times 10^{-19} \text{ s}) \times 0.998654545 \\ \times 0.999999999303923 \approx 8.71109 \times 10^{-19} \text{ s} \quad (6)$$

4. Resolving the Hubble Tension via Localized Clock Drift

The Hubble Tension is an artifact of **Timestamp Desynchronization**. Cosmic expansion (H_0) represents the isotropic frequency of the central server's

synchronization wave (Ping) . A local physical ping measurement of 73 km/s/Mpc converts directly to the server's clean background update frequency:

$$H_0 = 73 / (3.08567758 \times 10^{19}) \approx 2.366 \times 10^{-18} \text{ Hz} \quad (7)$$

Because the terrestrial observer operates within a nested local processing lag ($F_{local_Earth} < 1$), the local reference clock drifts relative to the global baseline. When observing the ancient Cosmic Microwave Background (CMB) across vast distances (unburdened WAN), the data timestamp is offset, producing the apparent, lower expansion rate of ~ 67 km/s/Mpc [5]. The tension is not cosmological expansion changing speed, but a standard network Jitter between a local and global server query observed from a lagged terminal.

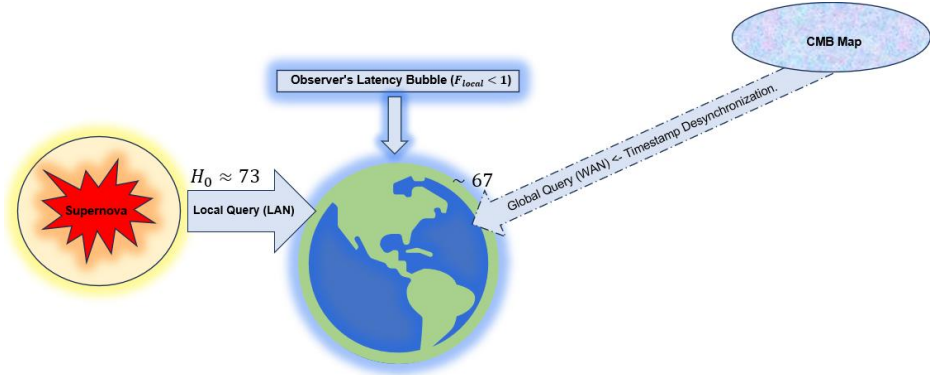


Figure 1: The Observer's Latency Bubble and Timestamp Desynchronization. A schematic representation of the Hubble Tension as an algorithmic network artifact. Nearby astronomical observations (e.g., Supernovae) function as a low-latency Local Area Network (LAN) query, yielding the clean server ping of $H_0 \approx 73$ km/s/Mpc. Conversely, ancient Cosmic Microwave Background (CMB) data packets travel vast cosmic distances (WAN query) and encounter the observer's local

processing latency bubble ($F_{local} < 1$) on Earth. This mismatch causes a systematic timestamp desynchronization (Jitter), shifting the perceived global expansion rate down to $\sim 67 \text{ km/s/Mpc}$.

5. Micro-Macro "Lazy Evaluation" Protocols

To maintain processing efficiency and prevent hardware crashes, the Base OS applies algorithmic **Lazy Evaluation** across both microscopic and macroscopic domains.

5.1 Micro-Scale: Quantum Superposition as Unrendered State Space

In quantum mechanics, particles are modeled as occupying infinite states simultaneously. In this framework, superposition is a resource-conservation protocol. The server does not allocate specific coordinates on the NavMesh for unobserved particles to avoid systemic bandwidth overload. They are kept in a computationally inexpensive probabilistic state (the wave function). Wave-function collapse is a definitive "System Call" (Audit) forced by localized physical collisions, requiring the engine to compute and instantiate a rigid coordinate.

The Attosecond Pulse Apparent Violation: State-of-the-art attosecond pump-probe spectroscopy measures light pulses as short as 0.4 as [6], which is nominally less than the local tick rate ($R_{conso_Earth} \approx 0.871 \text{ as}$). This is not a violation of the server clock, but an interference phase effect. Because the photons are maintained in superposition (probabilistic mathematical state in the Backend), the server computes the wave mathematics at high internal resolution, allowing destructive interference to yield sub-tick macro-structural phase differences. However, once a physical detector registers the interaction (the Frontend GUI), the data is forced through the local 0.871 as update filter,

displaying the 0.4 as duration as a mathematical phase aliasing artifact rather than a physical point-particle update.

5.2 Macro-Scale: Galactic Rotation and the Floating-Point Noise Floor (a_0)

At the outer edges of galaxies, gravitational acceleration (a) drops significantly. When the required spatial update vectors become too small, calculating individual trajectories threatens a floating-point overflow. To prevent this, the Base OS attempts to trigger a "Snap to Grid" Lazy Evaluation once local acceleration drops to the absolute noise floor of the system (a_0) [3]:

$$a_0 \approx \frac{c \cdot H_0}{2\pi} \approx \frac{299,792,458 \times 2.366 \times 10^{-18}}{2\pi} \approx 1.128 \times 10^{-10} \text{ m/s}^2 \quad (8)$$

Historically, modified dynamics models assumed individual particle gravity vectors are simply ignored below this absolute threshold. However, our framework dictates that this macroscopic lock—where outer stars are bound to the constant rotational drag of the local NavMesh without requiring Dark Matter—is not universal. It is a strictly conditional protocol requiring structural data integrity.

5.3 The Geometric Trigger: Z-Axis Culling and the Effective Noise Floor

While a_0 represents the base floating-point noise floor, macro-scale Lazy Evaluation is strictly gated by the geometric flatness of the structure.

In computational rendering, calculating three-dimensional continuous gravitational vectors requires exponentially more bandwidth than lateral, planar gravity (Z-Axis culling). To successfully lock a data structure onto the Golden Ratio grid without coordinate overlap, the mass payload must be highly compressed along the vertical axis.

We define the measurable physical flatness of a galaxy using the axis ratio $\eta = b/a$ (where b is vertical thickness and a is radial diameter). A perfect sphere yields $\eta = 1$, while a perfectly flat disk yields $\eta \rightarrow 0$. We introduce the **Geometric Flattening Factor** as $\sqrt{1 - \eta^2}$, which acts as a definitive mathematical gate for the evaluation threshold. The system's **Effective Noise Floor** is therefore modified:

$$a_{0_effective} = a_0 \times \sqrt{1 - \eta^2}$$

The Computational Lockdown:

For a standard, highly flattened spiral galaxy ($\eta \rightarrow 0$), the geometric factor converges to 1. Thus, $a_{0_effective} \approx a_0$. When the local kinematic acceleration drops to this threshold, the planar data structure achieves the optimal length required to preserve Golden Ratio integrity. The Base OS executes the "Snap-to-Grid" protocol, freezing the structure laterally and preserving rotation without requiring further continuous vector updates.

Conversely, for spherical, ultra-diffuse, or chaotic dwarf galaxies (e.g., NGC 1052-DF2), the geometry is highly uncompressed ($\eta \rightarrow 1$). The geometric factor drops toward zero, effectively reducing the Lazy Evaluation threshold to zero ($a_{0_effective} = 0$). Because the system's radial acceleration (a) cannot reach an absolute zero, the Base OS cannot safely lock the data structure, disabling the Lazy Evaluation trigger entirely. The system is forced to maintain continuous Newtonian calculations, which elegantly explains why these specific spherical galaxies exhibit absolute natural rotational decay and zero Dark Matter effects.

The Unified Lazy Evaluation Trigger

To summarize the macro-scale algorithmic protocol, we can mathematically synthesize the global server ping (H_0), the maximum transmission bandwidth (c), the geometric radial dispersion (2π), and the Z-Axis structural flattening requirement (η) into a single, unified equation. The definitive threshold for the system's Effective Noise Floor is formalized as:

$$a_{0_effective} = \frac{c \cdot H_0}{2\pi} \times \sqrt{1 - \eta^2}$$

This equation is not merely a gravitational limit, but an algorithmic phase transition boundary. When a local system's kinetic acceleration decays to this exact mathematical threshold, and provided the physical matter is flattened ($\eta \rightarrow 0$), the Base OS initiates a structural lock. The galaxy transitions into the server's lowest possible computational energy state—the Golden Ratio (ϕ) orbital packing.

By locking the architecture into this ϕ -optimized configuration, the universe eliminates the need for continuous, energy-intensive vector updates, resolving the requirement for Dark Matter and exposing flat galactic rotation curves as the ultimate signature of cosmic algorithmic efficiency.

Ultimately, Dark Matter is not a missing physical particle, but a manifestation of the universe's inherent **Law of Information Conservation**—packing kinematic data into the most computationally efficient geometric state (the ϕ ratio) when continuous rendering is no longer sustainable.

Milky Way

$$\text{Ratio } \eta = b/a \approx 0.01$$

$$\sqrt{1 - \eta^2} \approx 0.9999$$

$$a_0 \approx \frac{c \cdot H_0}{2\pi} \times \sqrt{1 - \eta^2} \approx 1.12794 \times 10^{-10}$$

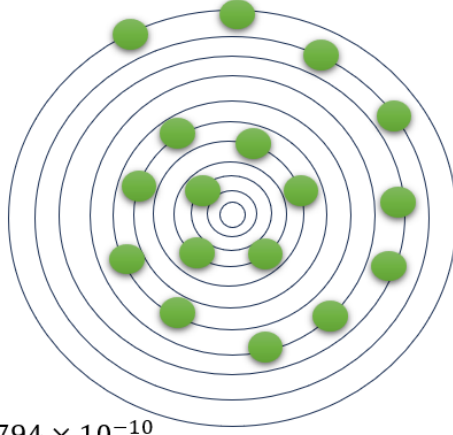


Figure 2: Macro-Scale "Snap-to-Grid" Protocol on the NavMesh.

A visual representation of algorithmic Lazy Evaluation in a flattened spiral galaxy. Below the effective noise floor ($a_{0_effective}$), individual kinematic vector tracking is suspended. Star nodes (green) are locked onto the discrete concentric processing tracks of the spatial grid. Governed by Golden Ratio (ϕ) packing geometry, the entire structure rotates synchronously as a single macro-object, producing flat rotation curves without the need for Dark Matter particles.

6. The Falsifiable Challenge: Earth vs. ISS Quantum Delay

To prove that physical gravity and motion are integrated computational latency artifacts of the observer, we propose a direct, empirical experiment.

Recent quantum optics experiments demonstrate "negative group delay" where a photon traversing a dense atomic cloud registers an apparent negative transit time (timestamp drift) [7]. Because our Consolidated Processing Rate (R_{conso}) accounts for both gravitational overhead and kinematic translation velocity, the total rendering lag differs predictably between Earth and orbit:

1. Earth's Surface Observer:

At typical laboratory latitudes, the gravitational latency dominates, while rotational velocity is negligible ($v \approx 300$ m/s), yielding:

$$\begin{aligned} F_{local_Earth} &\approx F_{grav_Earth} \times F_{kin_Earth} \\ &\approx (1 - 6.96 \times 10^{-10}) \times (1 - 5 \times 10^{-13}) \\ &\approx 1 - 6.96 \times 10^{-10} \end{aligned}$$

2. ISS Observer:

At an orbital altitude of ~ 400 km and a high orbital velocity of $v \approx 7.66$ km/s, the kinetic bandwidth depletion from coordinate translation exceeds the gravitational latency saved by altitude:

$$\begin{aligned} F_{local_ISS} &\approx F_{grav_ISS} \times F_{kin_ISS} \\ &\approx (1 - 6.55 \times 10^{-10}) \times (1 - 3.26 \times 10^{-10}) \\ &\approx 1 - 9.81 \times 10^{-10} \end{aligned}$$

Quantitative Prediction:

Standard physics claims that the local laboratory's motion and gravitational potential have zero effect on internal quantum atomic excitation transition delays.

In contrast, our framework predicts that the measured "negative time" timestamp drift ($\Delta T_{apparent}$) of the photon exiting the atomic cloud on the ISS

will be shifted toward a higher delay compared to Earth's surface, corresponding exactly to the nested rendering ratio:

$$\Delta T_{\text{apparent_ISS}} = \Delta T_{\text{apparent_Earth}} \times \left(\frac{F_{\text{local_ISS}}}{F_{\text{local_Earth}}} \right) \quad (9)$$

A verified shift matching this ratio will empirically prove that spacetime coordinates are updated on a discrete, resource-constrained network, uniting quantum mechanics and relativity under a single Base Operating System.

We challenge the global experimental community to run this test. Any measurable, systematic drift in quantum transition delays matching Equation (9) across different gravitational potentials will empirically falsify continuous spacetime, proving that physical reality is a nested, rendered digital interface.

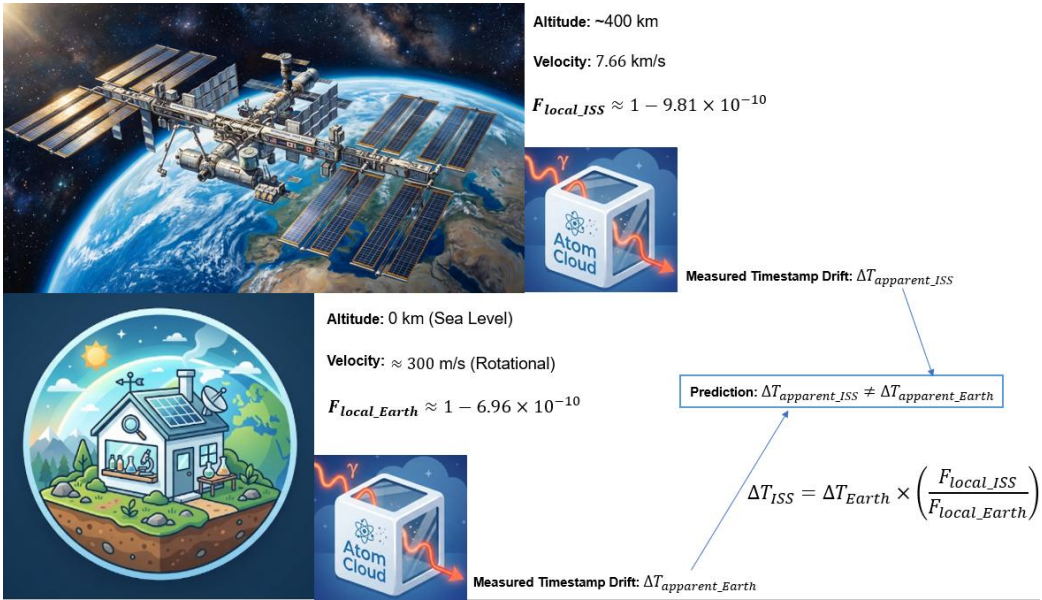


Figure 3: The Nested Rendering Test (Earth vs. ISS Quantum Delay). Experimental setup and quantitative predictions for the proposed quantum optics challenge. (Bottom) The terrestrial observer operates under high gravitational latency but negligible kinematic speed,

resulting in $F_{local_Earth} \approx 1 - 6.96 \times 10^{-10}$. (Top) The ISS observer saves gravitational latency due to altitude, but heavily drains the server's update bandwidth due to its extreme orbital speed ($v \approx 7.66$ km/s), yielding a higher overall latency of $F_{local_ISS} \approx 1 - 9.81 \times 10^{-10}$. The model predicts a precise, measurable shift in the quantum 'negative group delay' (ΔT) between the two platforms, dictated by the nested rendering ratio: $\Delta T_{apparent_ISS} = \Delta T_{apparent_Earth} \times (F_{local_ISS} / F_{local_Earth})$.

References

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